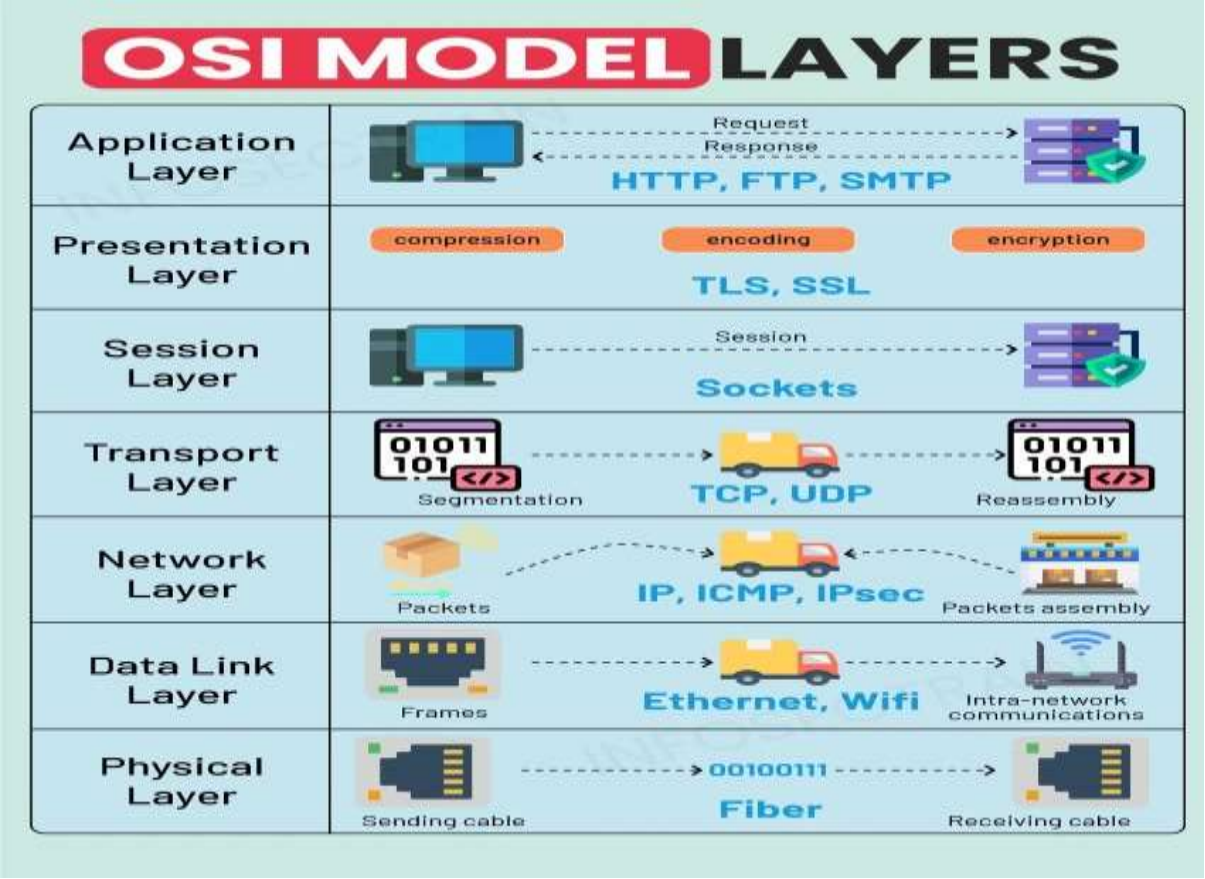
OSI LAYER

TASK-1 Draw a diagram of the OSI 7-layer model and label each layer with its correct name and a real-world example of a device or protocol that operates at that layer (e.g., Switch at Data Link, HTTP at Application).



1.PHISCAL – ETHERNET CABLE
2.DATA LINK- SWITCH
3.NETWORK- ROUTER
4.TRANSPORT-TCP
5.SESSION-NETBIOS
6.PRESENTIONS-SSL/TLS
7.APPLICATION-HTTP/HTTPS

OSI LAYER

TASK-2 Match the following networking scenarios to the correct OSI layer: a) Encrypting data before sending, b) Routing packets across networks, c) Converting data to electrical signals, d) Opening a web page in your.

Scenario	Correct OSI Layer	Layer Number
a) Encrypting data before sending	Presentation	Layer 6
b) Routing packets across networks	Network	Layer 3
c) Converting data to electrical signals	Physical	Layer 1
d) Opening a web page in your browser	Application	Layer 7

OSI LAYER

TASK-3 Pick any app you use daily (like WhatsApp, Instagram, or Zomato) and describe how at least three different OSI layers are involved when you send or receive data in that app.

1. Application Layer (Layer 7): WhatsApp lets me type and send a message to my friend.
2. Transport Layer (Layer 4): TCP or other transport mechanisms help transfer data between devices and servers reliably where applicable.
3. Network Layer (Layer 3): IP addresses help route data packets across networks to the destination.
4. Data Link Layer (Layer 2): Wi-Fi or mobile network technology transfers frames across the local network link.
5. Physical Layer (Layer 1): Radio signals or cables carry the data as bits.

OSI LAYER

TASK-4 Given this scenario: You are unable to access a website, but you can successfully ping its IP address. Identify which OSI layer(s) might be causing the problem and explain your reasoning.

Possible OSI Layers: Application (Layer 7) and Transport (Layer 4).

Reasoning:

1. Application Layer (Layer 7): The website or web server application may be down, or there may be a problem with DNS name resolution.
2. Transport Layer (Layer 4): HTTP/HTTPS traffic may be blocked by a firewall, or the required TCP port (80 or 443) may be unavailable.